

Analysis of Contemporary Economic Issues

Guidance 5: The European Green Deal (Part 2)

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**SOLVAY BRUSSELS SCHOOL
ECONOMICS & MANAGEMENT**

Outline

- 1 The EU Emissions Trading System (ETS) & CBAM
- 2 Just Transition & Energy
- 3 Transport & Taxation
- 4 Industry & Sustainable Finance
- 5 Biodiversity, Digital & Context
- 6 Impact Assessments (Belgium Focus)

The EU Emissions Trading System (ETS)

Mechanism

Puts a price on carbon and lowers the cap on emissions from certain economic sectors every year.

Track Record:

- Successfully brought down emissions from power generation and energy-intensive industries by **42.8%** in the past 16 years.

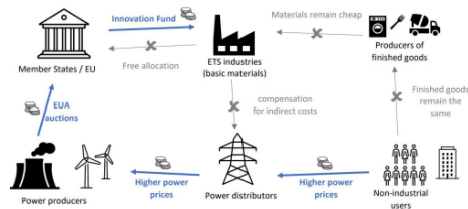
Commission Proposals:

- Lower the overall emission cap even further + increase annual reduction rate.
- Phase out free emission allowances.
- Include **shipping emissions** for the first time.
- Set up a **separate new ETS** for fuel distribution, road transport and buildings.

The Use of ETS Revenues

Allocation of Revenues:

- Member States must spend revenues on **climate and energy-related projects**.
- **Social Impact:** Part of revenues (from the new system for transport/buildings) addresses impact on vulnerable households, micro-enterprises, and transport users.



Flow of ETS Revenues & Allowances

The Effort Sharing Regulation

Definition

Assigns strengthened emissions reduction targets to each Member State for sectors such as buildings, road/domestic maritime transport, agriculture, waste, small industries.

Fairness Principle:

- Recognizes different starting points and capacities.
- Targets are based on **GDP per capita**.
- Adjustments made to take cost efficiency into account.

ETS and ESR are complementary:

- **ETS**: Covers direct, market-based emissions from large industries.
- **ESR**: Mandates binding, national-level reduction targets for non-traded sectors.

The Carbon Border Adjustment Mechanism (CBAM)

The Problem: Carbon Leakage

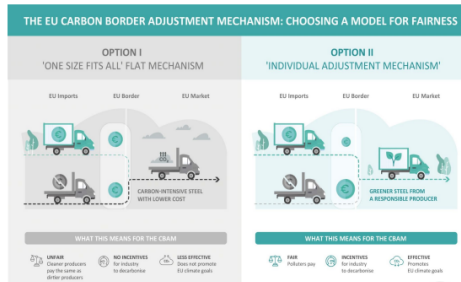
- Countries don't share the same ambition.
- Risk: Production transferred to countries with lower carbon costs or EU products replaced by carbon-intensive products.
- Result: No reduction in global emissions, thus frustrating EU industries efforts.

The Solution (CBAM):

- Puts a carbon price on imports of targeted products.
- Aims to encourage industry outside the EU to reduce their carbon use.
- Ensures domestic production and imports are subject to similar carbon pricing levels.

Implementation of CBAM

- **Certificates System:** Importers purchase certificates to cover embedded emissions.
- **Pricing:** Price calculated based on the weekly average auction price of EU ETS allowances (€/tonne CO_2).
- **Phasing:** Introduced gradually. Initially applies only to goods at **high risk of carbon leakage** (e.g., steel, cement).
- **Equal Treatment:** Importers pay the same carbon price as domestic producers under the ETS.
- **Revenue:** Potential EU "own resource" contributing to the EU budget.



Drawbacks and Impacts of CBAM

Drawbacks:

① Resource Shuffling:

- Exporters might send their "cleanest" production to the EU and "dirty" production elsewhere.
- Overall carbon intensity in the home market remains constant.

② Competitiveness of Downstream Industries:

- EU industries using inputs from CBAM sectors (e.g., a car maker buying steel) face higher costs compared with non-EU competitors which do not implement carbon pricing.

Impacts:

- **Benefits in terms of emission reductions and reduction of carbon leakage risks.**
- **Economic impacts:** Limited negative impacts on GDP and consumption, slightly positive effects on investments.
- **Social impacts:** Small increase in employments in sectors covered by CBAM, limited negative impact on employment for downstream sectors.
- **Administrative and compliance costs:** For business and authorities.

The Just Transition Mechanism (JTM)

Motto: "Making Sure No One Is Left Behind".

- **Goal:** Ensure the transition to climate-neutrality happens in a fair way by providing targeted support to alleviate the socio-economic impact of the transition.
- **Budget:** Mobilise at least **€55 billion** (2021-2027).
- **Target:** Most affected regions and sectors (high dependence on fossil fuels).



JTM: Who Benefits and How?

Citizens

- Employment opportunities in new sectors.
- Re-skilling.
- Energy-efficient housing.
- Fight energy poverty.
- Access to clean, affordable and secure energy.

Companies

- Transition to low-carbon tech.
- Economic diversification.
- Attractive conditions for investors.
- Loans/Financial support.
- Research and innovation.

Member States

- Green economy jobs.
- Sustainable transport.
- Renewable energy sources.
- Local public authority loans.
- Energy infrastructure.

Promoting Greener Energy

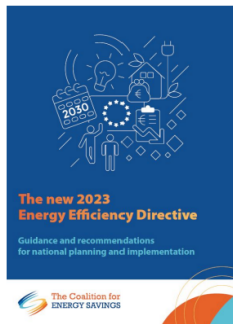
Energy production and use accounts for 75% of EU emissions.

Renewable Energy Directive

- **Target:** 40% of EU energy from renewable sources by 2030.
- Specific targets for transport, heating, cooling, buildings, industry.
- All member states will contribute to this goal.
- Specific targets for renewable energy use in transport, heating and cooling, buildings and industry.

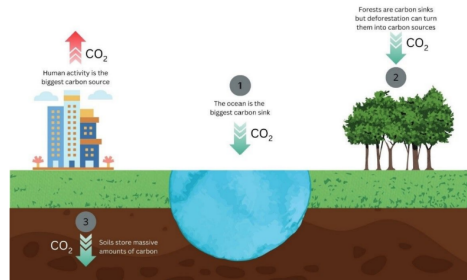
Energy Efficiency Directive

- **Binding Annual Target:** More ambitious target for reducing energy use at EU level.
- **National Contributions:** Almost doubles the annual energy saving obligation for Member States.
- **Public Sector Role:** Required to renovate **3%** of its buildings each year (Renovation Wave).
- **Goals:** Create jobs, bring down energy use, reduce costs for taxpayers.



Land Use, Forestry and Agriculture

- **Carbon Sinks:** Sets an overall EU target for carbon removals by natural sinks.
- **2035 Goal:** Climate neutrality in the land use, forestry, and agriculture sectors.
- **EU Forest Strategy:**
 - Plant **3 billion trees** by 2030.
 - Improve quality/resilience of forests.
 - Preserve biodiversity.



Forests: Sinks vs Sources

Road Transport Evolution

Emissions Standards

- **2030:** -55% average emissions for new cars/vans (vs 2021).
- **2035:** -100% (Zero-emission mobility).

Alternative Fuels Infrastructure Regulation:

- Expand charging capacity inline with sales.
- **Electric Charging:** Every 60 km on major highways.
- **Hydrogen Refuelling:** Every 150 km.



Tech comparison: BEV (Battery) vs Fuel Cell (Hydrogen).

Aviation and Maritime Transport

- **Infrastructure:** Aircraft and ships must have access to clean electricity supply in major ports/airports.
- **ReFuelEU Aviation:** Obliges fuel suppliers to blend increasing levels of sustainable aviation fuels in jet fuel.
- **FuelEU Maritime:**
 - Stimulate uptake of sustainable maritime fuels.
 - Set maximum limit on GHG content of energy used by ships calling at EU ports.

Energy Taxation Directive (ETD): Problems vs Objectives

Current Problems

- **Fossil Subsidies Persistence:** Divergent national rates and exemptions.
- **Misalignment:** Not aligned with ETS or Green Deal goals. Neither with GHG reduction or energy efficiency goals.
- **Obsolescence:** Minimum tax rates have lost their effect.

Revision Objectives

- **Alignment:** Taxation based on energy content and environmental performance (not volume).
- **Broadening Base:** Removing exemptions (e.g., Kerosene for aviation, Heavy oil for maritime).
- **Internal Market:** Preserving proper functioning.

The Net-Zero Industry Act

Will strengthen resilience and competitiveness of net-zero industries by creating better conditions for them and attracting investments.

Aim: Net-zero tech manufacturing capacity → **40%** of EU deployment needs by 2030.

Main Pillars:

- 1 **Investment Conditions:** Reduce administrative burden, simplify permits.
- 2 **CO2 Capture:** 50Mt injection capacity by 2030 (Oil/Gas contribution).
- 3 **Market Access:** Sustainability as a criteria in public procurement.
- 4 **Skills:** Academies for skilled workforce.
- 5 **Innovation:** Regulatory sandboxes to test technologies.

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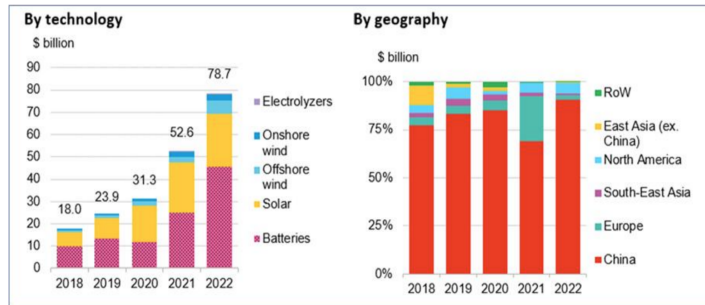


Figure 1: Investments in manufacturing of clean energy technologies. Source: Bloomberg NEF.

Figure 1 Analysis

- China dominates clean energy technologies manufacturing.
- Europe is one of the lowest in 2022, possibly due to supply chain shortages.

The Net-Zero Industry Act

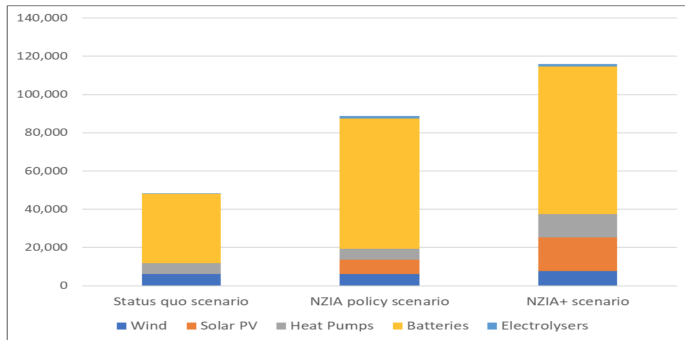


Figure 4: Investment needs to expand EU net-zero technology manufacturing capacity according to different scenarios (EUR million, cumulative 2023-2030)

Figure 1 Analysis

- **Status quo:** Current market shares of EU manufacturing technologies remain the same up to 2030.
- **NZIA policy scenario:** EU manufacturing market shares of NZ technologies are boosted.
- **NZIA+:** EU manufacturing capacity meets all EU demand for NZ technologies

Investment Needs (Fit-for-55)

Investment Gap (Annual):

- **Total Energy & Transport:** +€477 Billion/year needed compared to 2011-2020 trend.
- **Supply Side:** +€93 Billion (Power plants, grids).
- **Demand Side:** +€178 Billion (Residential, Tertiary, Industry).

Conclusion: Future capacity is lacking; massive acceleration needed.

Annex 1: Fit-for-55 and REPowerEU deployment investment

Table 9: Average annual investment needs in the energy system and for transport, historical trend 2011-2020, and Fit-for-55 policy scenario 2021-2030, (EUR 2022, billion) (annual)⁶⁴

Sector	2011-2020 (annual)	Fit-for-55 policy scenario 2021-30 (annual)	Difference (annual)
Supply side	55	148	+93
Power grid	15	55	+40
Power plants, incl. boilers and new fuels	40	93	+53
Demand side	160	339	+178
Industrial sector	12	34	+22
Residential	102	202	+100
Tertiary	46	103	+56
Total (Energy System)	215	487	+272
Transport sector ⁶⁵	549	754	+205
Total (energy and transport)	764	1,241	+477

Investments: Historical vs Fit-for-55 Scenario

What is the EU Taxonomy?

A classification system establishing a list of environmentally sustainable economic activities.

Criteria for "Environmentally Sustainable":

- Substantial contribution to at least one objective (Climate mitigation, Adaptation, Biodiversity, Circular economy, Pollution, Water).
- Does not harm any other objective.
- Meet minimum social safeguards.

Benefits: Prevents greenwashing, helps transition planning, lowers cost of capital for green projects.

Biodiversity Strategy & Nature Restoration Law

- **Interdependence:** Biodiversity loss and climate change are linked.
- Essential for achieving climate objectives, for life and for business.

Biodiversity Strategy Goals:

- Protected areas: $\geq 30\%$ of land/seas.
- Reduce pesticides by 50%.
- Restore 25,000 km of rivers.
- Unlock €20 billion/year.

Nature Restoration Law (Targets):

- **2030:** Restoration measures in 20% of EU land/sea areas.
- **2050:** Measures for *all* ecosystems in need.

Digital Transition & Green Deal

- **Europe** must strengthen its digital sovereignty and set standards.
- **Digital Services Act:** Standard for accountability of online platforms, protecting users.

Twin Challenges: Neither can succeed without the other.

- **Pros:** Digital reduces footprint (smart farming, videoconferencing).
- **Cons:** Energy consumption of digital tech (8-10% of consumption).

Objective for Data Centres (by 2030):

- 1 Climate Neutral.
- 2 Energy Efficient.
- 3 Sustainable.

What has changed in 2025 : Omnibus Package

Context of simplification:

- Fewer sectors covered by obligations.
- Pushed back timelines for compliance and reporting.
- Fewer data required.
- Exemptions added and simplified requirements.

Impact:

- Harder to meet targets.
- Higher risk of emission lock-in.
- Greater uncertainty for planning transition investments.
- Credibility at risk.

Regulatory Impact Assessments

Definition: Assessing positive/negative effects of regulations before adoption.

- Evidence-based policies and predictable environment are needed to encourage these investments.
- Impact assessments are not intended to replace political by technocratic decision making. Decisions will remain political but with better information.

Advantages

- Better use of policy instruments at all levels of government.
- Enhance credibility and transparency by explaining to stakeholders and general public.
- Helps to identify best options.
- Facilitate the convergence between political decision makers at all levels.
- Limited transparency.

State of Play in Belgium

Composite indicators' values of regulatory policy and governance practices published by the OECD have decreased for Belgium while the EU and OECD average moved in the opposite direction.

Factual elements confirm this:

- Quantitative impacts on GHG emissions of the federal energy and climate plan are only available for a minority of measures included (e.g no data on social impacts).
- Important policy measures are adopted without impact assessment and/or dropped without evaluation (e.g rents of poorly insulated housing).
- No systematic publication of impact assessment reports.
- And so on ...

Recommendations for Belgium

How to mainstream Impact Assessments?

- ① **Engagement:** Promote stakeholder engagement early.
- ② **Transparency:** Web portals at each government level.
- ③ **Scope:** Clarify which initiatives require assessments.
- ④ **Quality:**
 - Guidelines, toolboxes, training.
 - Mobilise research community (scientific input).
 - Develop quantification models.
- ⑤ **Process:** Begin early in the political cycle + systematic quality control.

Conclusion

- EU has reduced emissions while increasing GDP, but future targets are much more demanding.
- Transition requires significant reallocation of resources and massive investment (+€477bn/year).
- **Conditions for success:**
 - Wide range of policy instruments (ETS, CBAM, Taxonomy, etc.).
 - Clear link between biodiversity and climate.
 - Evidence-based decision making (Impact Assessments).
- **Belgium:** Margin for progress in providing evidence base for the transition.

Any Questions?

Clicker Questions

1. (UA) Please select the fully accurate statement concerning the Carbon Border Adjustment Mechanism (CBAM).

- A. It aims to reduce carbon leakage by imposing a carbon price on exports from the European Union.
- B. It may negatively affect the competitiveness of downstream EU industries that use inputs from CBAM-covered sectors.
- C. Upon its launch, it strictly and immediately applies to all goods imported into the European Union.
- D. It ensures a reduction of the overall carbon intensity in the exporting country's domestic market by preventing resource shuffling.

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2. (MA) Please select the fully accurate statements concerning the Emissions Trading System (ETS) and the Effort Sharing Regulation (ESR).

- A. The European Commission proposes to set up a separate new ETS for fuel distribution, road transport, and buildings.
- B. The Effort Sharing Regulation (ESR) sets binding reduction targets for Member States based solely on their population size.
- C. The ETS and ESR are complementary: the ESR covers sectors not subject to the industrial carbon market, such as agriculture, waste, and domestic transport.
- D. Member States are free to use ETS revenues for any public expenditure to reduce their national deficit.

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3. (UA) Please select the fully accurate statement concerning industrial sovereignty and investment needs in the green transition.

- A. The Net-Zero Industry Act sets a target for the EU to manufacture 100% of its deployment needs in net-zero technologies by 2030.
- B. China controls by far the largest share of global manufacturing and critical material refining capacity for many clean energy technology supply chains.
- C. To achieve the Fit-for-55 targets, annual investments in energy and transport will need to decrease compared to the 2011-2020 trend.
- D. In 2022, Europe ranked as the world's largest investor in clean energy technology manufacturing.

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4. (MA) Please select the fully accurate statements concerning the economic, social, and regulatory conditions of the European Green Deal.

- A. According to the EU Taxonomy, an economic activity is considered sustainable if it substantially contributes to at least one environmental objective while doing no significant harm to any other.
- B. The revision of the Energy Taxation Directive (ETD) aims to base taxation on the physical volume of fuels rather than their energy content.
- C. Regarding Regulatory Impact Assessments (RIAs), Belgium shows excellent results and ranked well above the OECD average in 2021.
- D. The Just Transition Mechanism (JTM) aims to alleviate the socio-economic impact of the transition by providing targeted support to regions and sectors most dependent on fossil fuels.

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